

1. Administrative & Study Identification

Full Study Title: A Clinical Study of MK-1084 With Targeted Therapy and Chemotherapy in People With Colorectal Cancer (MK-1084-012/KANDLELIT-012) **Short Title/Acronym:** KANDLELIT-012 / MK-1084-012 **Protocol Number:** U1111-1311-8311 / MK-1084-012 **NCT Number:** NCT06997497 **Sponsor Name:** Merck **Company:** Merck **Trial Status:** RECRUITING **Investigational Product:** MK-1084 (calderasib) **Phase of Development:** PHASE3 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: In Part 1, to evaluate the safety and tolerability of MK-1084 with cetuximab and mFOLFOX6. In Part 2, the primary objective is to evaluate Progression-Free Survival (PFS) in participants receiving MK-1084 plus cetuximab and mFOLFOX6 compared to standard care. **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Duration of Response (DOR), Overall Survival (OS), and overall safety profile.

2.2 Background & Rationale Researchers are looking for other ways to treat locally advanced or metastatic colorectal cancer (mCRC) that is unresectable and has a gene mutation called KRAS G12C. Standard (or usual) treatments for this type of colorectal cancer may include mFOLFOX6 with or without bevacizumab. This study aims to determine if adding MK-1084 (the study medicine) and cetuximab to mFOLFOX6 can treat locally advanced or mCRC with the KRAS G12C mutation. MK-1084 and cetuximab are targeted therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (mFOLFOX6 with or without bevacizumab) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Targeted Enrollment: 477 participants **Number of Sites:** ~133 global locations (including USA, Argentina, Australia, Brazil, Canada, Chile, and China) **Estimated Study Start:** 16-Jul-2025 **Estimated Primary Completion:** 27-Aug-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a histologically confirmed diagnosis of locally advanced unresectable or metastatic (unresectable Stage III or Stage IV as defined by American Joint Committee on Cancer [AJCC] eighth edition) colorectal adenocarcinoma. 2. Part 2 only: Has not received systemic anticancer therapy for locally advanced unresectable or metastatic colorectal cancer. 3. Tumor tissue demonstrates presence of a Kirsten rat sarcoma viral oncogene homolog G12C (KRAS G12C) mutation. 4. Human immunodeficiency virus (HIV)-infected participants must have well controlled HIV on antiretroviral therapy (ART). 5. Participants who are Hepatitis B surface antigen (HBsAg) positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy and have undetectable HBV viral load. 6. Participants with history of hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable.

4.2 Exclusion Criteria 1. Has active inflammatory bowel disease requiring immunosuppressive medication or previous clear history of inflammatory bowel disease (eg, Crohn's disease, ulcerative colitis, chronic diarrhea). 2. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease. 3. Has known dihydropyrimidine dehydrogenase (DPD) deficiency. 4. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 5. Has received prior systemic anticancer therapy including investigational agents within 4 weeks before randomization. 6. Has 1 or more conditions that, in the opinion of the investigator, make the participant ineligible for treatment with bevacizumab. 7. Has known additional malignancy that is progressing or has required active treatment within the past 3 years. 8. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis or leptomeningeal disease. 9. Has active infection requiring systemic therapy. 10. Has not adequately recovered from major surgery or have ongoing surgical complications. 11. Has a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-1084 given orally; cetuximab given intravenously (IV); mFOLFOX6 (oxaliplatin, leucovorin, 5-fluorouracil, calcium) given IV. **Route of Administration:** Oral and Intravenous **Duration of Treatment:** Until disease progression, unacceptable toxicity, or study withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-related toxicities (e.g.,

antiemetics). **Prohibited:** Other investigational agents or prior systemic anticancer therapies within 4 weeks of randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Mutational Testing (KRAS G12C), Baseline Imaging
Baseline	Day 0	Randomization, First Dose, Safety Labs
Interim	Every 2 Weeks	Safety Labs, Efficacy Assessment, IV Infusions (mFOLFOX6/Cetuximab)
End of Study	At Progression	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

1. Number of Participants Experiencing Dose-Limiting Toxicity (DLT): A DLT is defined as the occurrence of protocol-specified toxicities if assessed by the investigator to be possibly, probably, or definitely related to study intervention administration.
2. Part 1: Number of Participants Who Experience an Adverse Event (AE): An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention.
3. Part 1: Number of Participants Who Discontinue Study Treatment Due to an AE.

7.2 Secondary & Exploratory Endpoints

1. Objective Response Rate (ORR)
2. Overall Survival (OS)
3. Duration of Response (DOR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via patient questionnaires, HCP evaluation, and clinical lab monitoring.

Serious Adverse Events (SAEs): Standard reporting timelines, typically within 24 hours of investigator awareness.

Data Monitoring Committee (DMC): External Data Monitoring Committee to oversee safety data periodically throughout the study.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy measures; Safety Set for all participants who received at least one dose of study medication.

Sample Size Justification: Approximately 477 participants to detect a statistically significant difference in outcome measures.

Handling of Missing Data: Standard censoring rules applied for time-to-event endpoints (OS, DOR) if patients drop out or are lost to follow-up.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and centralized remote monitoring visits by the sponsor. **Audit Trail:** Validated 21 CFR Part 11 compliant electronic data capture (EDC) system with strict access control and eCRF locking mechanisms.

11. Study Identifiers & Governance

Posting ID: 516746307885808500 **Primary ID:** MK-1084-012 **Registry Number:** NCT06997497 **Sponsor/Lead Organization:** Merck **Study Title:** KANDLELIT-012 **Official Regulatory Title:** A Phase 3, Randomized, Open-label, Multicenter Clinical Study to Evaluate the Safety and Efficacy of MK-1084, Cetuximab, and mFOLFOX6 Versus mFOLFOX6 With or Without Bevacizumab as First-line Treatment of Participants With KRAS G12C-mutant, Locally Advanced Unresectable or Metastatic Colorectal Cancer (KANDLELIT-012)

12. Clinical Framework (Oncology Specific)

Primary Condition: Colorectal Neoplasms **Disease Areas:** Colon Adenocarcinoma, Rectal Adenocarcinoma **Diagnosis Threshold:** Histologically confirmed locally advanced unresectable or metastatic colorectal adenocarcinoma with KRAS G12C mutation. **Medical Methodology:** Targeted therapy + Standard Chemotherapy (mFOLFOX6) **Associated Drugs:** calcium

13. Study Procedures & Methodology

Management Pathway: First-line systemic therapy for KRAS G12C mutated mCRC **Education Protocol (HCP):** Protocol-specific training and adverse event management for targeted therapy + chemo. **Education Protocol (Patient):** Informed consent process, dosing diaries, and self-monitoring guidance for oral drug (MK-1084) adherence. **Quality Audit Mechanism:** Centralized imaging review (BICR) to track and verify tumor response parameters.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up for overall survival spanning multiple years until study completion (Est. 2030) **Onsite Visit Cadence:** Bi-weekly (Q2W) visits to align with standard IV chemotherapy infusion cycles. **Remote Monitoring Frequency:** Intermittent remote visits and safety check-ins as defined by site SOPs. **Checklist**

Review Interval: Regular interval scans (approx. every 8-12 weeks) for RECIST v1.1 evaluation.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					